Consensus-Guided Prompt Selection for Test-Time Adaptation of Vision-Language Models

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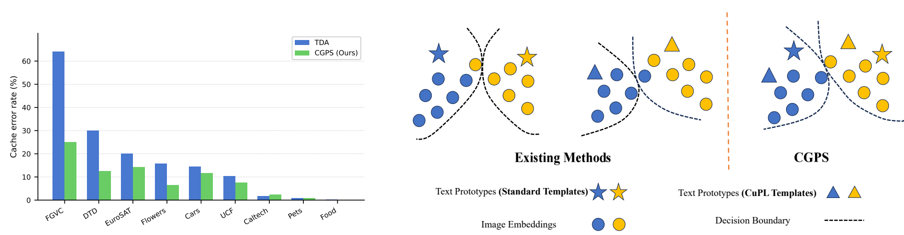
Abstract. Vision-language models (VLMs) achieve strong zero-shot recognition but degrade under distribution shift, motivating test-time adaptation (TTA) without target-domain supervision. However, existing cache-based and distribution-based TTA methods share a common weakness: they both rely on a single text classifier to judge sample reliability. Under domain shift, this classifier is poorly calibrated and often assigns high confidence to wrong predictions. These mislabelled samples accumulate over time, corrupting visual caches and biasing distribution estimates. To address this problem, we propose Consensus-Guided Prompt Selection (CGPS), a training-free method that addresses this weakness through two complementary modules. The Consensus-Guided Accumulator (CGA) accepts a sample only when two classifiers built from different prompt vocabularies agree. It accumulates per-class visual centroids from these reliable samples. The Visual-Guided Prompt Selector (VGPS) scores each candidate prompt against these centroids and selects those most discriminative in the target domain, forming a refined text classifier. Experiments on ten cross-domain benchmarks show that CGPS improves state-of-the-art methods on average and integrates seamlessly into existing TTA frameworks at negligible cost.

Keywords: Test-time adaptation · Vision-language models · CLIP

1 Introduction

Vision-language foundation models have demonstrated remarkable zero-shot understanding across a wide range of visual tasks by training on web-scale image-text pairs [21]. Taking CLIP as a representative example, it enables zero-shot image classification through predefined text prompts, without requiring any labeled target data. Despite this broad capability, performance degrades noticeably when the test distribution diverges from the pretraining data [11, 14]. Fine-tuning on the target domain can recover accuracy but demands labeled data and non-trivial computational effort, which limits its practicality in many deployment settings.

Test-time adaptation (TTA) addresses this by adapting model behavior at inference time using only the incoming unlabeled test stream, without touching the pretrained weights [23]. Two families of training-free TTA have become prominent for VLMs. Cache-based methods, such as TDA [12], BoostAdapter [30], and DPE [27], maintain a dynamic store of test image features alongside their predicted labels. A sample enters the cache only when the text classifier assigns it a low-entropy prediction, treating entropy as a proxy for reliability. At inference, predictions combine zero-shot text logits with cache-retrieval logits to incorporate accumulated visual evidence. Distribution-based methods, such as DOTA [9] and FreeTTA [4], model each class as a Gaussian distribution refined by online expectation-maximization (EM). The text classifier first assigns soft labels to incoming images. The class mean and covariance are then updated by those soft labels, allowing the model to track the target visual distribution.



(a) (b)

Fig. 1. (a) Cache error rates across cross-domain benchmarks. Entropy-based method (TDA) admits large amounts of incorrect samples on domain-shifted datasets, our consensus filter consistently reduces cache contamination. (b) Feature space visualization: existing methods produce a misaligned decision boundary under domain shift, CGPS uses the consensus of both classifiers to determine a more reliable boundary.

Both method families, however, place full trust in a single text classifier to decide which samples are reliable. CLIP's text encoder is unreliable on out-of-distribution inputs, and high-confidence predictions frequently carry incorrect labels, as we observe empirically in Fig. 1(a). For cache-based methods, this causes the cache to accumulate mislabelled entries that corrupt the stored class prototypes. For distribution-based methods, incorrect soft labels propagate into every EM update, biasing the estimated class distributions. As shown in Fig. 1(a), TDA's cache consistently contains a high proportion of wrong labels across datasets, showing that a single text classifier cannot reliably judge sample quality under domain shift. The underlying cause is geometric: domain shift reduces the alignment between visual feature clusters and their corresponding text prototypes, so the decision boundary produced by a single text classifier becomes unreliable, as illustrated in Fig. 1(b).

To address this, we propose Consensus-Guided Prompt Selection (CGPS), a training-free method designed to alleviate the reliability limitations of existing VLM TTA frameworks. CGPS consists of two components: a Consensus-Guided Accumulator (CGA) and a Visual-Guided Prompt Selector (VGPS). Specifically, CGA filters test samples by requiring agreement between two classifiers constructed from entirely different prompt vocabularies, and accumulates reliable per-class visual centroids from the agreed samples. Based on these centroids, VGPS evaluates each candidate description by its alignment with the target visual distribution, retaining the most domain-relevant prompts to form a refined text classifier.

Our contributions are as follows.

- We propose CGPS, a plug-in method applicable to existing mainstream VLM TTA frameworks to improve adaptation reliability under domain shift.
- We introduce CGA, which filters test samples via dual-template consensus, mitigating the incorrect sample admission inherent in single-classifier strategies.
- We introduce VGPS, which uses the accumulated visual centroids to identify discriminative prompts, constructing a refined text classifier better suited to the target domain.
- Experiments on diverse benchmarks validate the effectiveness of CGPS, improving state-of-the-art TTA approaches on average.

2 Related Work

2.1 Vision-Language Models

Large-scale vision-language models (VLMs) such as CLIP [21] are pre-trained on web-scale image-text pairs via contrastive learning, enabling zero-shot image classification through text prompts without task-specific training. Numerous efforts have focused on adapting these models to downstream tasks [29, 26]. CoOp [32] optimizes continuous prompt vectors with a small set of labeled examples, while CoCoOp [31] conditions them on individual image features for better cross-domain generalization. Despite their effectiveness in few-shot settings, these methods require labeled target-domain data, limiting applicability when annotations are unavailable. CGPS sidesteps this requirement entirely, adapting the text classifier online from unlabeled test samples alone, without supervision or gradient updates.

2.2 Vision-Language Test-Time Adaptation

Test-time adaptation (TTA) for VLMs improves generalization to unseen target domains using only unlabeled test samples at inference time [23]. Recent approaches fall into two main categories: cache-based and distribution-based methods. Cache-based methods build a dynamic memory of test features to refine predictions without backpropagation. TDA [12] constructs positive and negative feature caches from test samples and fuses cache-retrieval logits with zero-shot predictions. BoostAdapter [30] augments the cache with a sample-adaptive regional bootstrapping strategy for more precise memory construction.

DPE [27] jointly evolves visual and textual prototypes incrementally to capture domain-specific semantics over the test stream. Distribution-based methods model the test stream as class-conditional Gaussians updated via online expectation-maximization. DOTA [9] estimates class-conditional test distributions and computes posteriors from the learned distributions for adaptation. BCA [33] updates both class embeddings and class priors under a Bayesian view of VLM prediction. FreeTTA [4] proposes a training-free online EM procedure for target distribution modeling. ReTA [15] combines consistency-aware entropy reweighting with text-embedding distribution calibration to improve reliability under visual variations. All these methods, however, rely on a single text classifier's entropy or confidence to make reliability decisions. Under distribution shift, CLIP frequently assigns high confidence to incorrect predictions (Fig. 1(a)), making this signal unreliable. CGPS addresses this weakness by requiring agreement between two classifiers built from different prompt vocabularies before admitting a sample as reliable. This consensus criterion reduces filter error and enables more trustworthy adaptation.

2.3 Test-Time Prompt Tuning

Zero-shot CLIP performance depends critically on text prompt design. CuPL [20] and DCLIP [17] enrich the text classifier with LLM-generated, per-class descriptive sentences, substantially improving accuracy on fine-grained datasets. However, LLM-generated descriptions vary widely in quality, and averaging the entire pool introduces domain-irrelevant semantics into the classifier. Gradient-based test-time prompt tuning adapts prompts at inference without labeled data. TPT [23] minimizes prediction entropy across augmented views of each test image, and DiffTPT [7] replaces standard augmentations with diffusion-generated views to increase view diversity. While effective, these methods require per-image backpropagation through the text encoder, imposing significant overhead unsuitable for streaming inference. CGPS offers a lightweight alternative that requires no gradient computation. It selects the most domain-relevant prompts from a fixed candidate pool by scoring each description against per-class visual centroids built from the test stream, adapting the text classifier without any backpropagation.

3 Method

CGPS consists of two components: the Consensus-Guided Accumulator and the Visual-Guided Prompt Selector. We first introduce the necessary background in Section 3.1, then present the Consensus-Guided Accumulator in Section 3.2, and describe the Visual-Guided Prompt Selector and its integration with existing TTA pipelines in Section 3.3. Figure 2 provides an overview of the full pipeline.

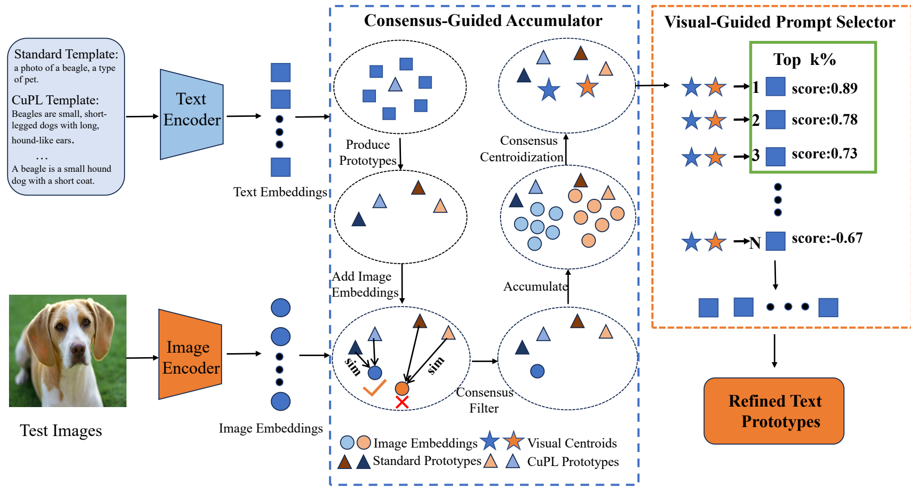


Fig. 2. Overview of the CGPS framework. Hand-crafted standard templates and LLM-generated CuPL templates are encoded by the text encoder to produce per-class standard prototypes and CuPL prototypes. Given a test image, its embedding is extracted by the image encoder and classified independently by both sets of prototypes. In the Consensus-Guided Accumulator, samples where the two prototype sets agree are added to the accumulator, and per-class consensus centroids are computed. In the Visual-Guided Prompt Selector, each prompt embedding is scored by its alignment with the visual centroid, and the highest-scoring prompts are selected to form the refined text prototype.

3.1 Preliminaries

Zero-Shot Classification with CLIP. CLIP [21] encodes images and text into a shared D-dimensional embedding space. For a C-class task, the predicted probability for class c is:

$$p_c = \frac{\exp(z^\top w_c/\tau)}{\sum_{j=1}^{C} \exp(z^\top w_j/\tau)}, \tag{1}$$

where z is the L2-normalized image embedding, $w_c \in \mathbb{R}^D$ is the text prototype for class c, τ is the temperature, and $\hat{y} = \arg\max_c p_c$.

Two Prompt Sources. We maintain two complementary text classifiers. Standard templates (`"a photo of a [class]"`) are encoded and averaged per class to produce $w_c^{\text{std}} \in \mathbb{R}^D$. CuPL prompts [20], per-class descriptions generated by a large language model, are averaged to produce $w_c^{\text{cupl}} \in \mathbb{R}^D$ with richer but variable-quality attribute coverage. Stacking all C prototypes forms classifier matrices W_{std} and $W_{\text{cupl}} \in \mathbb{R}^{C\times D}$, whose rows serve as w_c in Eq. (1). Relying on a single template source inevitably admits mislabelled samples. By requiring both classifiers to agree, CGPS selects only consensus samples and reduces such errors.

Online Visual Gaussian Estimation. DOTA [9] models each class as a Gaussian $\mathcal{N}(\mu_c^V, \Sigma_c)$ and updates its parameters online. The class mean μ_c^V and covariance Σ_c are updated as:

$$\mu_c^V \leftarrow \frac{n_c \mu_c^V + p_c z}{n_c + p_c}, \quad \Sigma_c \leftarrow \frac{n_c \Sigma_c + p_c (z - \mu_c^V)(z - \mu_c^V)^\top}{n_c + p_c}, \tag{2}$$

where n_c is the accumulated soft-label weight and p_c is the predicted probability for class c. A shared precision matrix Λ is derived from the class covariances:

$$\Lambda \; = \; \left[(1-\varepsilon)\,\bar{\Sigma} + \varepsilon I\right]^{-1}, \qquad \bar{\Sigma} = \frac{1}{C}\sum_{c=1}^{C} \Sigma_c, \tag{3}$$

where ε is a regularization coefficient and I is the identity matrix. The prediction score is:

$$s_c(x) \; = \; z^\top w_c^{\text{std}} + \gamma\big(z^\top \Lambda \mu_c^V - \tfrac{1}{2} {\mu_c^V}^\top \Lambda \mu_c^V\big), \tag{4}$$

where $\gamma = \min(\rho \cdot \bar{n}/B, \, \eta)$ is a weight that increases as more visual evidence accumulates, with $\bar{n}$ the mean per-class soft count and ρ, B, η constants from DOTA [9].

3.2 Consensus-Guided Accumulator

Existing TTA methods rely on a single text classifier to assess sample reliability. Cache-based methods such as TDA [12] gate cache admission by prediction entropy, while distribution-based methods such as DOTA [9] weight each M-step update by the corresponding softmax confidence. Both strategies break down under distribution shift, since CLIP's text encoder is not calibrated for out-of-distribution inputs and frequently assigns high confidence to incorrect predictions (Fig. 1(a)). The admitted samples thus carry a persistent fraction of labelling errors, which gradually bias cache prototypes in cache-based methods and corrupt the estimated visual distributions in distribution-based methods.

To address this, we propose the Consensus-Guided Accumulator, which identifies reliable samples by requiring agreement between two classifiers built from entirely different prompt vocabularies. This design is conceptually inspired by co-training [1], where complementary classifiers can reduce the risk of accepting erroneous pseudo-labels. Formally, we consider a test sample x to achieve consensus if the following three conditions hold simultaneously:

$$\hat{y}_{\text{std}}(x) = \hat{y}_{\text{cupl}}(x), \qquad p_{\text{std}}^{\max}(x) > \tau_c, \qquad p_{\text{cupl}}^{\max}(x) > \tau_c, \tag{5}$$

where $\hat{y}_{\text{std}}$ and $\hat{y}_{\text{cupl}}$ are the argmax predictions of W_{std} and W_{cupl}, $p^{\max}$ denotes the maximum class probability, and τ_c is a confidence threshold. Standard templates use concise, class-shared contexts, whereas CuPL prompts provide longer, class-specific, attribute-rich descriptions. Empirically, the two classifiers exhibit a low false-consensus rate: samples receiving consistent predictions reach an average error rate of 19.0%, substantially lower than 34.4% and 31.5% for the standard and CuPL classifiers alone. This confirms that dual-template consensus suppresses misclassified samples far more effectively than gating by a single classifier.

For each consensus sample with agreed prediction c, we maintain a per-class running sum $s_c \in \mathbb{R}^D$ and integer count m_c, both initialized to zero:

$$s_c \leftarrow s_c + z, \qquad m_c \leftarrow m_c + 1, \qquad \mu_c^{\mathrm{HC}} = s_c / m_c, \tag{6}$$

where $\mu_c^{\mathrm{HC}} \in \mathbb{R}^D$ is the visual centroid for class c, that is, the running mean of all consensus visual embeddings assigned to that class. This centroid accumulates in parallel with the base TTA method and serves as a reliable visual anchor for the Visual-Guided Prompt Selector in Section 3.3.

3.3 Visual-Guided Prompt Selector

While the Consensus-Guided Accumulator provides reliable per-class visual centroids, the CuPL description pool contains many candidates of uneven quality: some capture appearance attributes that are discriminative in the target domain, while others describe context or properties shared across visually similar classes. Averaging the entire pool dilutes the discriminative signal and introduces domain-irrelevant semantics into the text classifier. We therefore propose the Visual-Guided Prompt Selector, which evaluates each CuPL template against the visual evidence accumulated by the Consensus-Guided Accumulator and performs a one-shot selection at a fixed trigger point in the test stream.

At the trigger point, inspired by Fisher's linear discriminant analysis [8], we compute a per-class discriminant direction $d_c \in \mathbb{R}^D$ by whitening the consensus centroid with the precision matrix Λ from the visual Gaussian (Eq. (3)). Concretely, for each class c we compute:

$$d_c = \mathrm{normalize}\big(\Lambda\, \tilde{\mu}_c\big), \qquad \tilde{\mu}_c = \begin{cases} \mu_c^{\mathrm{HC}} & \text{if } m_c \geq m_0, \\ \mu_c^{V} & \text{otherwise} \end{cases}, \tag{7}$$

where m_0 is a minimum count threshold below which the visual centroid is considered unreliable, and μ_c^V is the visual class mean from the same Gaussian statistics as Λ, used as a fallback. We prefer μ_c^{HC} over μ_c^V as the primary anchor because it is built from consensus samples with very few labelling errors, and thus more faithfully reflects the true class centre.

Each CuPL template $t_{c,k}$ with L2-normalized text embedding $e_{c,k} \in \mathbb{R}^D$ is then scored by how well it points toward class c relative to all other classes:

$$s_{c,k} = e_{c,k}^{\top} d_c - \frac{1}{C-1} \sum_{c' \neq c} e_{c,k}^{\top} d_{c'}. \tag{8}$$

Algorithm 1 CGPS

Require: Test stream $\{x_t\}_{t=1}^{T}$; classifiers W_{std}, W_{cupl}; CuPL pool $\{\mathcal{P}^{c}_{\text{cupl}}\}$; base TTA method $\mathcal{A}$; trigger fraction ρ_{trig}; confidence threshold τ_c; minimum count m_0; selection fraction α.

```
Initialize s_c ← 0, m_c ← 0 for all c; refined←false.
for t = 1, ..., T do
    Encode x_t → z_t; compute ŷ_std, ŷ_cupl, p_std^max, p_cupl^max.
    if ŷ_std = ŷ_cupl and p_std^max > τ_c and p_cupl^max > τ_c then
        s_ŷ ← s_ŷ + z_t; m_ŷ ← m_ŷ + 1.
    end if
    if t ≥ ρ_trig T and not refined then
        Retrieve Λ from the visual Gaussian (Eq. (3)); compute d_c for all c via Eq. (7).
        Score e_{c,k} via Eq. (8); retain top-⌈αM_c⌉ per class.
        Form ŵ_c via Eq. (9); refined←true.
    end if
    Predict with A (substituting Ŵ as text classifier if refined); update A with (z_t, p_t).
end for
```

The first term measures how well template $t_{c,k}$ aligns with the discriminant direction of class c. The second term subtracts the average alignment with all other classes. A template that scores high must therefore describe appearance attributes specific to class c, not ones shared across many classes. We retain the top $K = \lceil \alpha M_c \rceil$ templates per class by score, where $\alpha \in (0, 1)$ is a selection fraction and M_c is the total number of CuPL templates for class c.

The refined prototype $\hat{w}_c$ combines the selected CuPL template embeddings with the standard-template prototype w_c^{std} as a stable anchor:

$$\hat{w}_c \;=\; \text{normalize}\left(\frac{1}{2}\, w_c^{\text{std}} + \frac{1}{2K} \sum_{k \in \text{top-}K} e_{c,k}\right). \tag{9}$$

The two components play complementary roles. w_c^{std} serves as a stable baseline that prevents the prototype from drifting when the selected CuPL templates are weakly discriminative, while the selected $e_{c,k}$ supply domain-specific visual detail that the standard templates lack. Equal weighting balances these two aspects. The result is normalized back onto the unit sphere, so $\hat{w}_c$ can directly replace any text prototype without modifying the prediction formula. If no CuPL template achieves a positive score ($\max_k s_{c,k} < 0$), we fall back to $\hat{w}_c = w_c^{\text{std}}$. The complete online procedure is given in Algorithm 1.

After selection, the new prototype replaces the original text classifier for the remaining predictions. For distribution-based methods such as DOTA, it replaces the text term in the prediction score, and the visual Gaussian continues to update under the same rule, now guided by pseudo-labels that reflect the new prototype. For cache-based methods such as TDA, it replaces the text term used in the classification logit and in the entropy that determines cache admission, while the cache's own image-based retrieval is unchanged. Because CGPS only modifies the text classifier, it integrates into either method family without changing any internal procedures.

4 Experiments

We describe the experimental setup including benchmark datasets, compared methods, and implementation details, then present results against representative TTA methods across three paradigms, followed by analyses of backbone generalization, hyperparameter sensitivity, component ablation, and computational efficiency.

4.1 Experimental Setup

Datasets. We evaluate on ten standard cross-domain benchmarks: Caltech-101 [6], DTD [3], Stanford Cars [13], EuroSAT [10], FGVC-Aircraft [16], Oxford Flowers [18], Oxford Pets [19], SUN397 [25], UCF-101 [24], and Food-101 [2]. They cover diverse visual domains including fine-grained recognition, satellite imagery, and action classification, providing a broad test of cross-domain generalization. All experiments apply a pre-trained CLIP directly to each target dataset without any target-domain supervision.

Compared Methods. We compare CGPS against methods from three paradigms. Prompt tuning: CoOp [32] and CoCoOp [31] learn continuous prompt vectors from labeled source-domain data, TPT [23] adapts prompts at test time via entropy minimization over augmented views, HisTPT [28] further incorporates historical test-sample knowledge banks. Cache-based adaptation: TDA [12] builds positive and negative feature caches gated by prediction entropy, BoostAdapter [30] improves cache quality through instance-adaptive regional bootstrapping. Distribution-aware adaptation: PromptAlign [22] aligns test statistics with source-domain distributions, BCA [33] updates class embeddings under a Bayesian framework, FreeTTA [4] models the target distribution via online EM, DOTA [9] estimates class-conditional visual Gaussians with online updates for adaptive prediction.

Implementation Details. We use CLIP with ViT-B/16 [21, 5] as the backbone, following the standard TTA setting with batch size 1, and report top-1 accuracy. For CGPS, we set the confidence threshold to 0.30, trigger fraction to 0.25, selection fraction to 0.10, and minimum consensus count to 5. The same configuration is used for all datasets without any per-dataset tuning.

4.2 Comparison with State-of-the-Art

Table 1 presents a systematic comparison across ten datasets with CLIP-ViT-B/16. The rows are arranged by paradigm, covering prompt tuning, cache-based adaptation, and distribution-aware adaptation.

Table 1. Top-1 accuracy (%) on ten cross-domain benchmarks with CLIP-ViT-B/16.

Method	Origin	Aircraft	Caltech101	Cars	DTD	EuroSAT	Flower102	Food101	Pets	SUN397	UCF101	Avg
Zero-Shot CLIP [21]	ICML'21	23.22	93.55	66.11	45.04	50.42	66.99	82.86	86.92	65.63	65.16	64.59
Prompt tuning												
CoOp [32]	IJCV'22	18.47	93.70	64.51	41.92	46.39	68.71	85.30	89.14	64.15	66.55	63.88
CoCoOp [31]	CVPR'22	22.29	93.79	64.90	45.45	39.23	70.85	83.97	**90.46**	66.89	68.44	64.63
TPT [23]	NeurIPS'22	24.78	94.16	66.87	47.75	42.44	68.98	84.67	87.79	65.50	68.04	65.10
HisTPT [28]	NeurIPS'24	**26.90**	**94.50**	**69.20**	**48.90**	**49.70**	**71.20**	**89.30**	89.10	**67.20**	**70.10**	**67.60**
Cache-based adaptation												
TDA [12]	CVPR'24	23.91	94.24	67.28	47.40	58.00	71.42	86.14	88.63	67.62	70.66	67.53
BoostAdapter [30]	NeurIPS'24	27.45	**94.77**	**69.30**	45.69	**61.22**	71.66	**87.17**	89.51	68.09	71.93	68.68
TDA+CGPS	Ours	**28.77**	94.48	67.27	**51.89**	60.96	**74.83**	85.83	**90.49**	**68.42**	**73.46**	**69.64**
Distribution-aware adaptation												
PromptAlign [22]	NeurIPS'23	24.80	94.01	68.50	47.24	47.86	72.39	86.65	90.76	67.54	69.47	66.92
BCA [33]	CVPR'25	28.59	**94.69**	66.86	53.49	56.63	73.12	85.97	90.43	68.41	67.59	68.59
FreeTTA [4]	CVPR'25	25.11	94.63	67.34	46.96	**62.93**	71.62	86.62	90.11	67.76	71.16	68.42
DOTA [9]	NeurIPS'25	26.25	94.16	**69.56**	47.64	62.78	75.23	87.08	92.01	69.80	72.54	69.71
DOTA+CGPS	Ours	**29.22**	**94.69**	69.53	**56.68**	62.40	**77.14**	**87.09**	**93.40**	**70.95**	**76.08**	**71.72**

On the ViT-B/16 backbone, CGPS improves both TDA and DOTA on average from a single fixed configuration. TDA+CGPS raises the TDA average from 67.53% to 69.64%, a gain of 2.11%. DOTA+CGPS raises the DOTA average from 69.71% to 71.72%, a gain of 2.01%.

Cache-Based Group. TDA improves the average to 67.53% by retaining reliable visual cache entries, while BoostAdapter advances to 68.68% through instance-adaptive bootstrapping. TDA+CGPS achieves 69.64%, with gains of 4.86% on Aircraft, 4.49% on DTD, 3.41% on Flowers, and 2.80% on UCF-101.

Distribution-Aware Group. BCA advances to 68.59% via Bayesian inference over class embedding posteriors. FreeTTA reaches 68.42% through online EM distribution modeling. DOTA achieves 69.71% through class-conditional visual Gaussian modeling with online updates. DOTA+CGPS further improves to 71.72%, a gain of 2.01%, confirming that consensus-driven text-classifier refinement provides complementary gains atop visual distribution modeling. The improvement is most pronounced on Aircraft by 2.97%, DTD by 9.04%, and UCF-101 by 3.54%. CGPS improves both TDA and DOTA on average across the ten benchmarks, demonstrating its applicability across different TTA paradigms.

4.3 Experimental Analysis

Backbone Generalization. To verify that CGPS generalises beyond ViT-B/16, we evaluate TDA+CGPS and DOTA+CGPS on five benchmarks with CLIP-RN50. As shown in Fig. 3, on DTD, TDA+CGPS gains 5.2% and DOTA+CGPS gains 8.4%. On Aircraft, the gains are 2.1% and 3.0% respectively. CGPS achieves consistent improvements across all five datasets, demonstrating that it generalises across different backbone architectures.

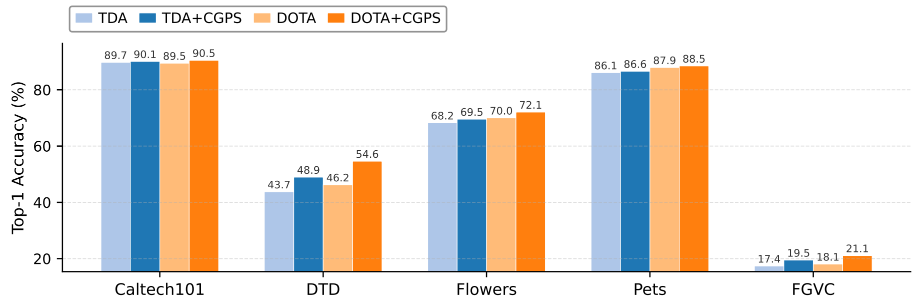


Fig. 3. Top-1 accuracy (%) on five benchmarks with CLIP-RN50.

Hyperparameter Sensitivity. We study the robustness of CGPS to its four hyperparameters: the trigger fraction ρ_{trig}, the top-template selection ratio α, the consensus confidence threshold τ_c, and the minimum consensus count m_0. For each hyperparameter, we vary it over a range of values while fixing the others at their defaults, and report accuracy of TDA+CGPS and DOTA+CGPS on DTD. As shown in Fig. 4, performance remains stable across all tested ranges, demonstrating that CGPS is robust to hyperparameter choices and works well with a single fixed configuration across all datasets.

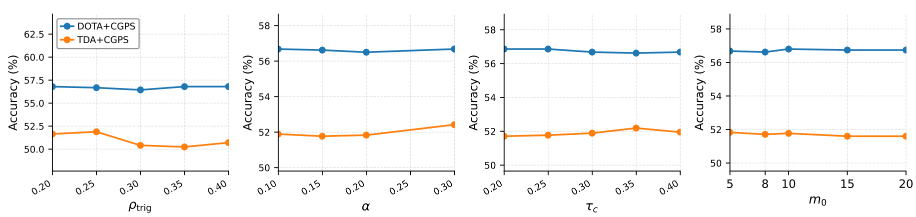


Fig. 4. Hyperparameter sensitivity analysis of CGPS on DTD with ViT-B/16.

Effectiveness of Each Component. We study the individual contributions of CGA and VGPS on both TDA and DOTA, with results shown in Table 2. CGA alone brings gains of 0.70% on TDA and 1.05% on DOTA. VGPS alone yields larger gains of 1.57% and 1.60%. CGPS, combining both components, achieves the best results with gains of 2.11% on TDA and 2.01% on DOTA. CGA filters samples via dual-classifier consensus, yielding cleaner per-class visual centroids that serve as more accurate anchors for prompt scoring. VGPS directly refines the text classifier with domain-discriminative prompts, but depends on accurate centroids to score them correctly. Without CGA, the centroids are noisier and prompt selection is less reliable. Together, the two modules are complementary:

combining both components achieves the best average accuracy on both TDA and DOTA.

Table 2. Component ablation on our datasets with ViT-B/16.

CGA	VGPS	TDA	Gain	DOTA	Gain
×	×	67.53	–	69.71	–
✓	×	68.23	+0.70	70.76	+1.05
×	✓	69.10	+1.57	71.31	+1.60
✓	✓	69.64	+2.11	71.72	+2.01

Computational Cost. We measure per-image adaptation time on a single NVIDIA RTX 4090 with ViT-B/16, averaged over 83,696 test images. Image-encoding time is excluded as it is shared across all methods. Results are shown in Table 3. DOTA+CGPS takes 3.46 ms per image, marginally above DOTA's 3.45 ms. TDA+CGPS takes 7.95 ms, adding only 0.40 ms over TDA's 7.55 ms. In contrast, BoostAdapter must encode 64 augmented views per sample, reaching 84 ms per image. CGPS performs prompt scoring only once during the entire test stream, so the overhead per image is negligible. It reuses visual centroids and text embeddings already available from the adaptation process, requiring no gradient computation or additional forward passes.

Table 3. Computational cost for CGPS on our datasets.

Method	ms/img	Average (%)
Zero-Shot CLIP	<0.1	64.59
TDA [12]	7.55	67.53
BoostAdapter [30]	84	68.68
TDA+CGPS	7.95	69.64
DOTA [9]	3.45	69.71
DOTA+CGPS	3.46	71.72

5 Conclusion

In this paper, we present CGPS, a training-free method for reliable VLM test-time adaptation. Existing TTA methods rely on a single text classifier to decide which samples are trustworthy. Under domain shift, this classifier is poorly calibrated and frequently assigns high confidence to wrong predictions, causing mislabelled samples to enter visual caches or bias distribution estimates. CGPS addresses this with two complementary components. The Consensus-Guided Accumulator requires agreement between two classifiers built from different prompt vocabularies. This filters out noisy samples and produces clean per-class visual centroids. The Visual-Guided Prompt Selector scores each candidate prompt against these centroids and retains those that best match the target domain, forming a more accurate text classifier. CGPS requires no retraining and adds little computational cost. It plugs into existing TTA pipelines without changing their visual adaptation steps. Experiments on ten cross-domain benchmarks show that CGPS improves state-of-the-art methods on average, raising TDA by 2.11% and DOTA by 2.01% on average. It also remains effective across different backbone architectures, confirming its value as a practical and reliable enhancement for test-time adaptation.

References

1. Blum, A., Mitchell, T.: Combining labeled and unlabeled data with co-training. In: Proceedings of the Annual Conference on Computational Learning Theory (COLT) (1998)
2. Bossard, L., Guillaumin, M., Van Gool, L.: Food-101 – mining discriminative components with random forests. In: European Conference on Computer Vision (ECCV) (2014)
3. Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., Vedaldi, A.: Describing textures in the wild. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2014)
4. Dai, Q., Yang, S.: Free on the fly: Enhancing flexibility in test-time adaptation with online EM. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2025)
5. Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. In: International Conference on Learning Representations (ICLR) (2021)
6. Fei-Fei, L., Fergus, R., Perona, P.: Learning generative visual models from few training examples: An incremental bayesian approach tested on 101 object categories. In: CVPR Workshop on Generative Model Based Vision (2004)
7. Feng, C.M., Yu, K., Liu, Y., Khan, S., Zuo, W.: Diverse data augmentation with diffusions for effective test-time prompt tuning. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2023)
8. Fisher, R.A.: The use of multiple measurements in taxonomic problems. Annals of Eugenics **7**(2), 179–188 (1936)
9. Han, Z., Yang, J., Wang, G., Li, J., Xu, Q., Shou, M.Z., Zhang, C.: DOTA: Distributional test-time adaptation of vision-language models. In: Advances in Neural Information Processing Systems (NeurIPS) (2025)
10. Helber, P., Bischke, B., Dengel, A., Borth, D.: EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (2019)
11. Hendrycks, D., Basart, S., Mu, N., et al.: The many faces of robustness: A critical analysis of out-of-distribution generalization. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2021)
12. Karmanov, A., Guan, D., Lu, S., El Saddik, A., Xing, E.: Efficient test-time adaptation of vision-language models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2024)

13. Krause, J., Stark, M., Deng, J., Fei-Fei, L.: 3D object representations for fine-grained categorization. In: ICCV Workshop on Fine-Grained Visual Recognition and Annotation (2013)
14. Liang, V.W., Zhang, Y., Kwon, Y., Yeung, S., Zou, J.Y.: Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning. In: Advances in Neural Information Processing Systems (NeurIPS) (2022)
15. Liang, Y., Chen, H., Xiong, Y., Zhou, Z., Lyu, M., Lin, Z., Niu, S., Zhao, S., Han, J., Ding, G.: Advancing reliable test-time adaptation of vision-language models under visual variations. In: Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM) (2025)
16. Maji, S., Rahtu, E., Kannala, J., Blaschko, M., Vedaldi, A.: Fine-grained visual classification of aircraft. arXiv preprint arXiv:1306.5151 (2013)
17. Menon, S., Vondrick, C.: Visual classification via description from large language models. In: International Conference on Learning Representations (ICLR) (2023)
18. Nilsback, M.E., Zisserman, A.: Automated flower classification over a large number of classes. In: Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) (2008)
19. Parkhi, O.M., Vedaldi, A., Zisserman, A., Jawahar, C.: Cats and dogs. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2012)
20. Pratt, S., Covert, I., Liu, R., Farhadi, A.: What does a platypus look like? generating customized prompts for zero-shot image classification. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2023)
21. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al.: Learning transferable visual models from natural language supervision. In: Proceedings of the International Conference on Machine Learning (ICML) (2021)
22. Samadh, J.H.A., Gani, H., Hussein, N.H., Khattak, M.U., Naseer, M., Khan, F.S., Khan, S.: Align your prompts: Test-time prompting with distribution alignment for zero-shot generalization. In: Advances in Neural Information Processing Systems (NeurIPS) (2023)
23. Shu, M., Nie, W., Huang, D.A., Yu, Z., Goldstein, T., Anandkumar, A., Xiao, C.: Test-time prompt tuning for zero-shot generalization in vision-language models. In: Advances in Neural Information Processing Systems (NeurIPS) (2022)
24. Soomro, K., Zamir, A.R., Shah, M.: UCF101: A dataset of 101 human actions classes from videos in the wild. Tech. rep., arXiv preprint arXiv:1212.0402 (2012)
25. Xiao, J., Hays, J., Ehinger, K.A., Oliva, A., Torralba, A.: SUN database: Large-scale scene recognition from abbey to zoo. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2010)
26. Yuan, H., Yu, N., Lu, Z., Huang, Y., Yang, J., Yan, J., Zhang, Z., Wang, L.: CAPNet: Context-aware prompt network for weakly-supervised open-world phrase-grounding. In: Proceedings of the International Conference on Image and Graphics (ICIG). pp. 187–199 (2025)
27. Zhang, C., Stepputtis, S., Sycara, K., Xie, Y.: Dual prototype evolving for test-time generalization of vision-language models. In: Advances in Neural Information Processing Systems (NeurIPS) (2024)
28. Zhang, J., Huang, J., Zhang, X., Shao, L., Lu, S.: Historical test-time prompt tuning for vision foundation models. In: Advances in Neural Information Processing Systems (NeurIPS) (2024)

29. Zhang, P., Li, Y., Wang, S.: RoICLIP: Text-enhanced UAV-based video object detection. In: Proceedings of the International Conference on Image and Graphics (ICIG). pp. 151–163 (2023)
30. Zhang, T., Wang, J., Guo, H., Dai, T., Chen, B., Xia, S.T.: BoostAdapter: Improving vision-language test-time adaptation via regional bootstrapping. In: Advances in Neural Information Processing Systems (NeurIPS) (2024)
31. Zhou, K., Yang, J., Loy, C.C., Liu, Z.: Conditional prompt learning for vision-language models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2022)
32. Zhou, K., Yang, J., Loy, C.C., Liu, Z.: Learning to prompt for vision-language models. International Journal of Computer Vision (IJCV) (2022)
33. Zhou, L., Ye, M., Li, S., Li, N., Zhu, X., Deng, L., Liu, H., Lei, Z.: Bayesian test-time adaptation for vision-language models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2025)